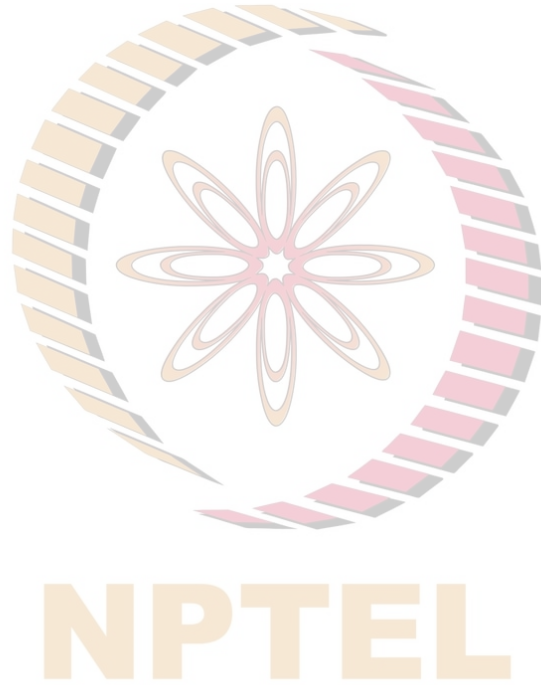
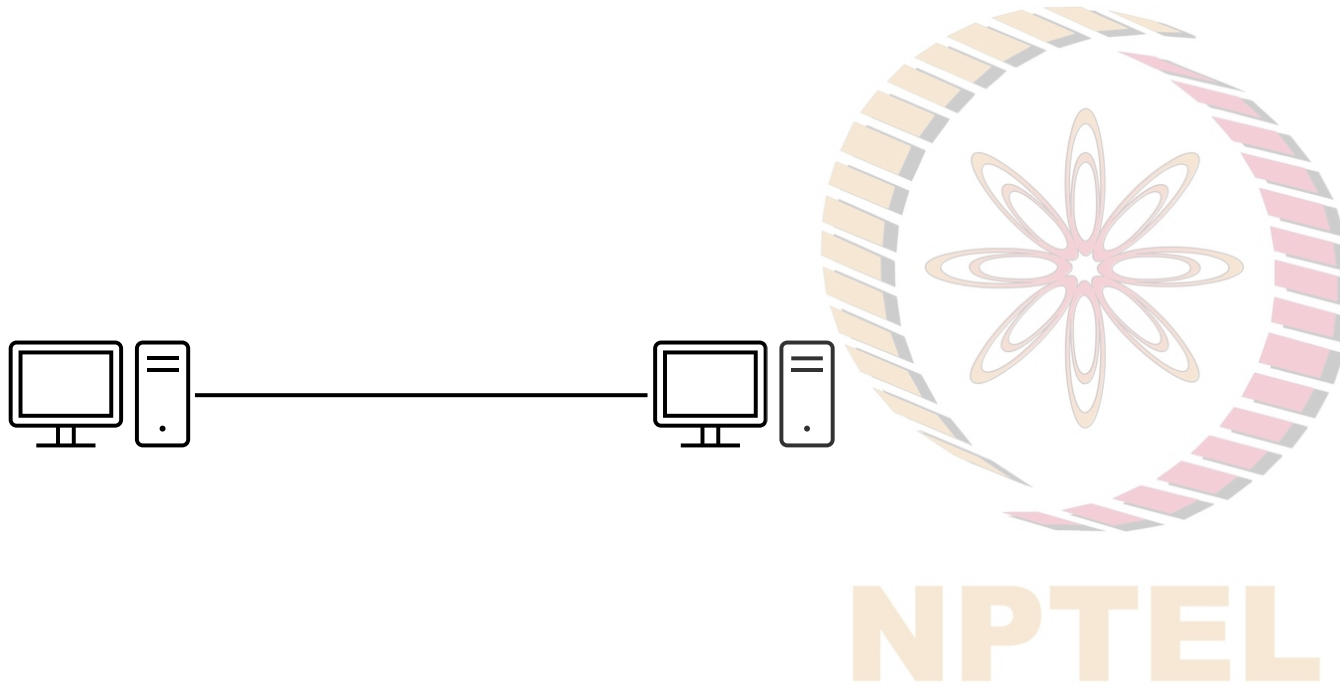


How Computers Can be Connected ?

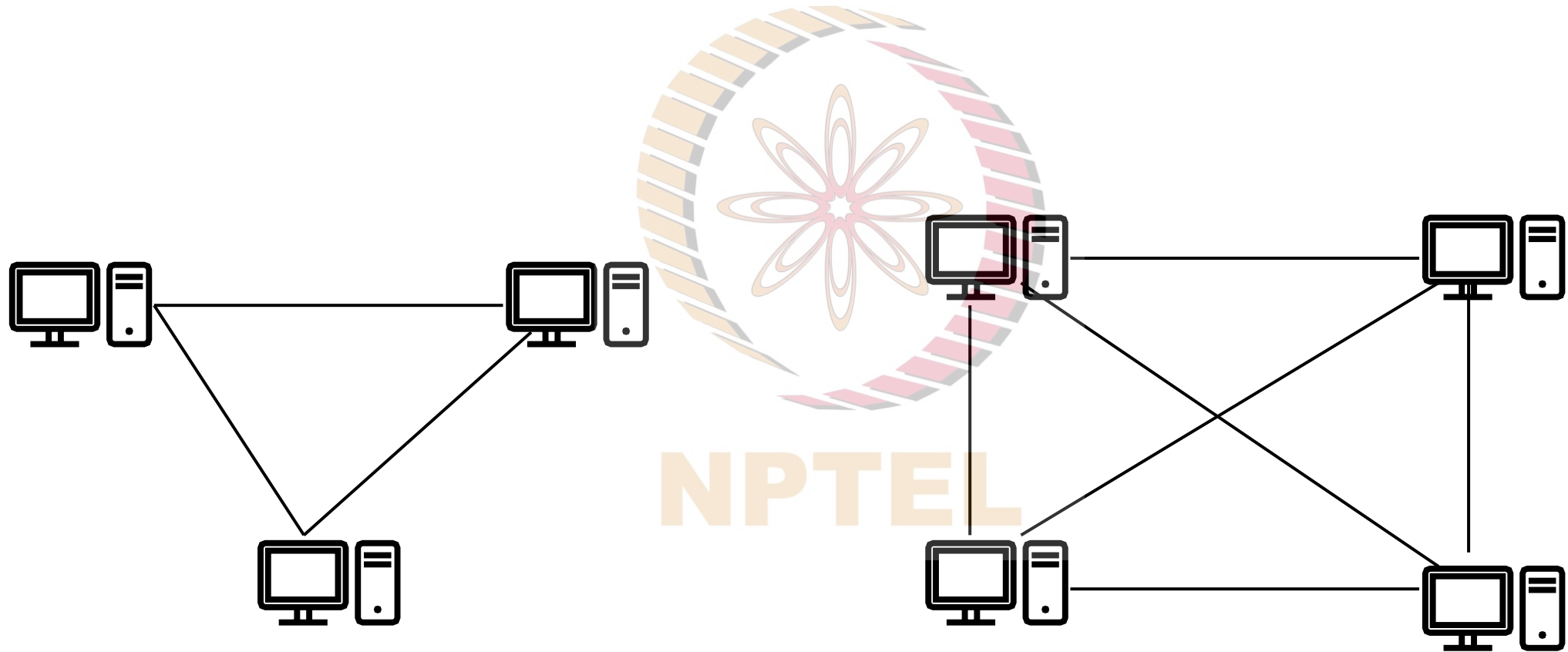
- ❑ Network topologies



How Computers Can be Connected ?



How Computers Can be Connected ?



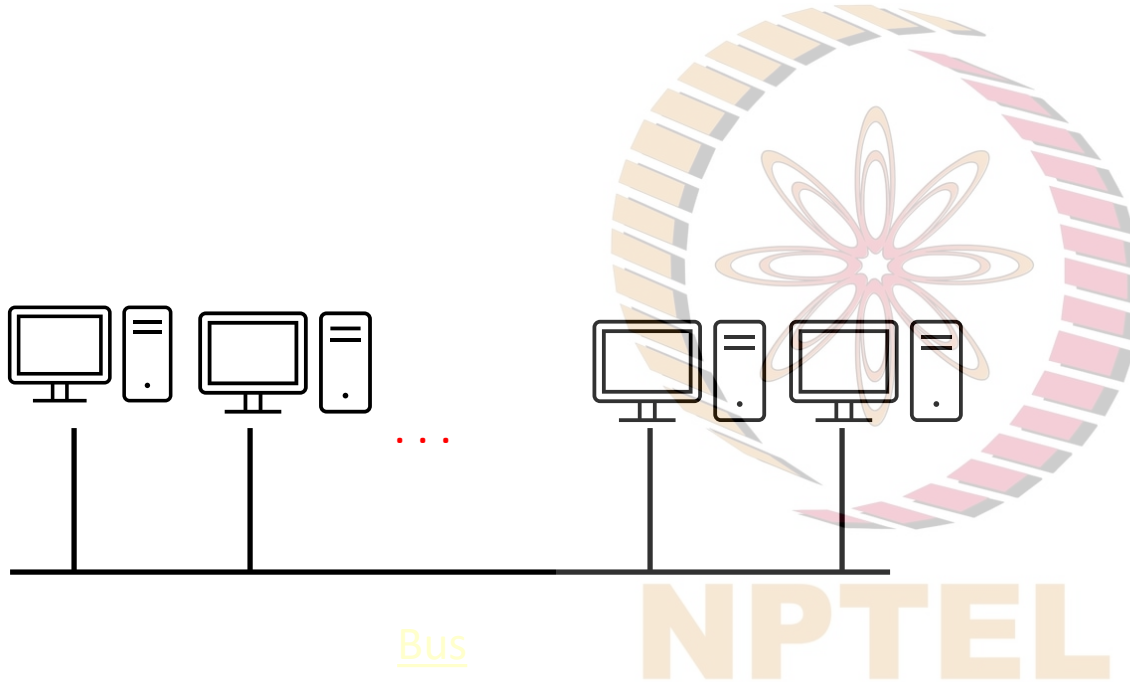
Point-to-Point Connectivity

- ❑ In this configuration we do not need addressing, routing, protocol, etc.
- ❑_Physical layer: coding, modulation, etc.



Connecting N users: *Directly*

□ *Bus*: broadcast, collisions, media access control



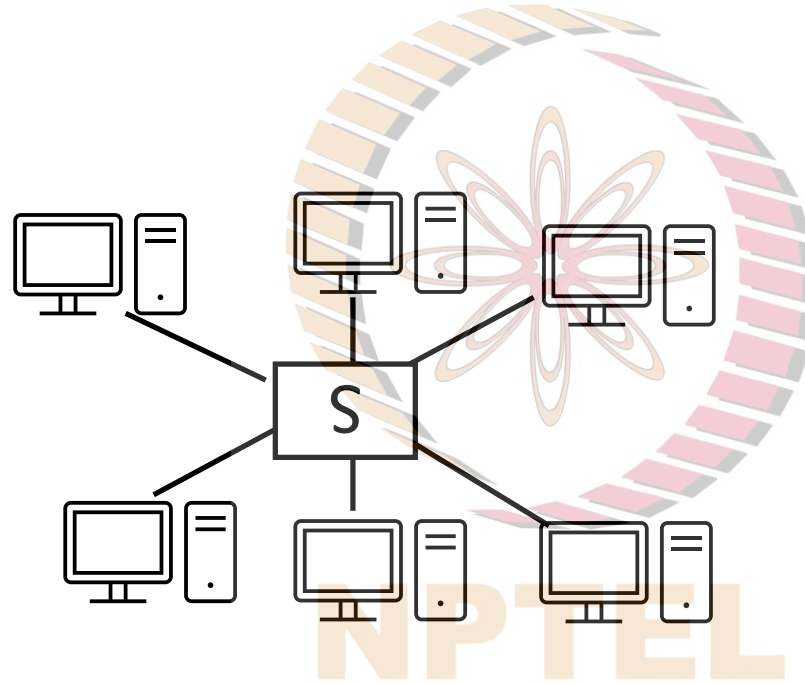
□ Address concept needed if we want the receiver *alone* to consume the data !

How to build Scalable Networks?

- ❑ Scaling: As number of computers connected increases, inefficiency increases.
 - ❑ Inefficiency limits scaling
- ❑ Direct connectivity is inefficient & hence does not scale
 - ❑ Mesh: *inefficient* in terms of # of links
 - ❑ Bus architecture: 1 expensive link, N cheap links. *Inefficient* in bandwidth use

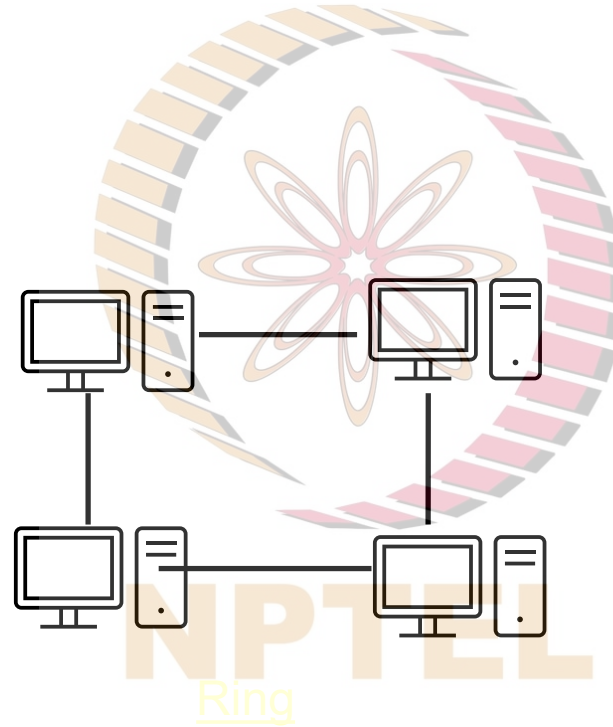
Connecting N Computers: *Indirectly*

❑ Star Topology

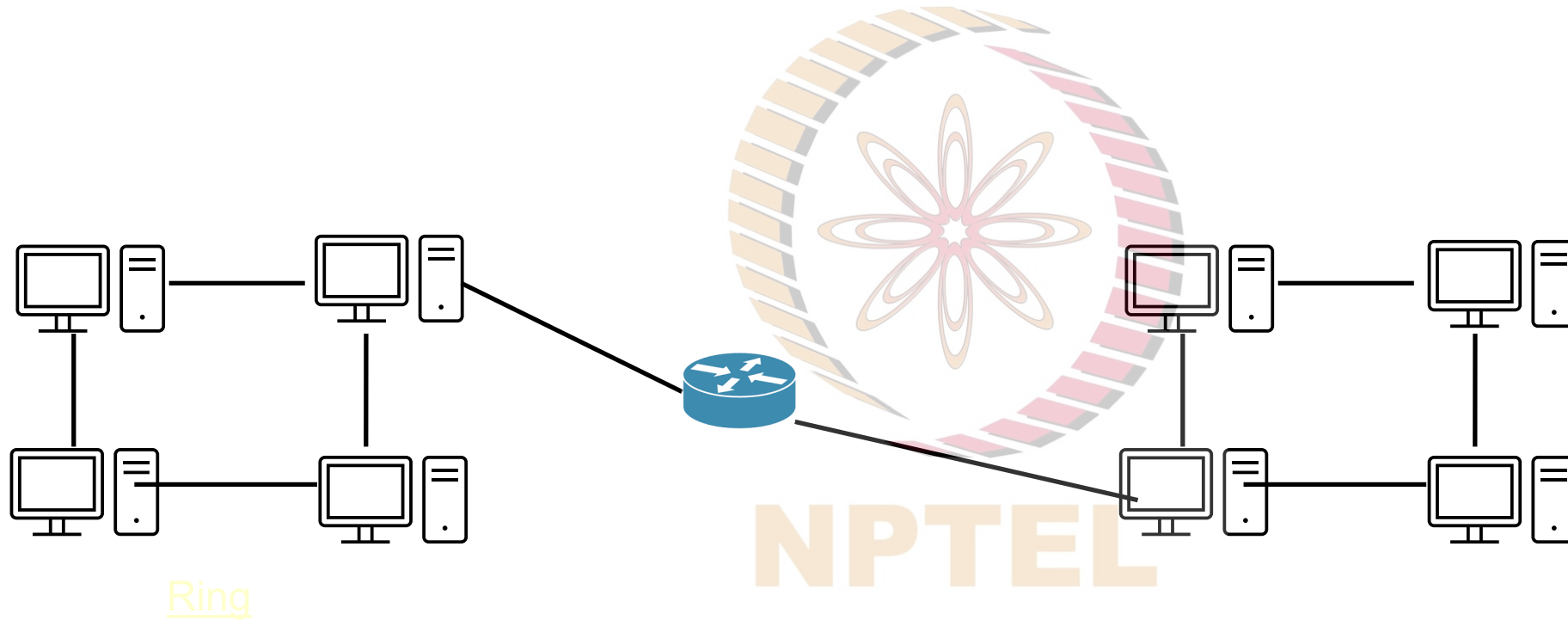


Connecting N users: *Indirectly*

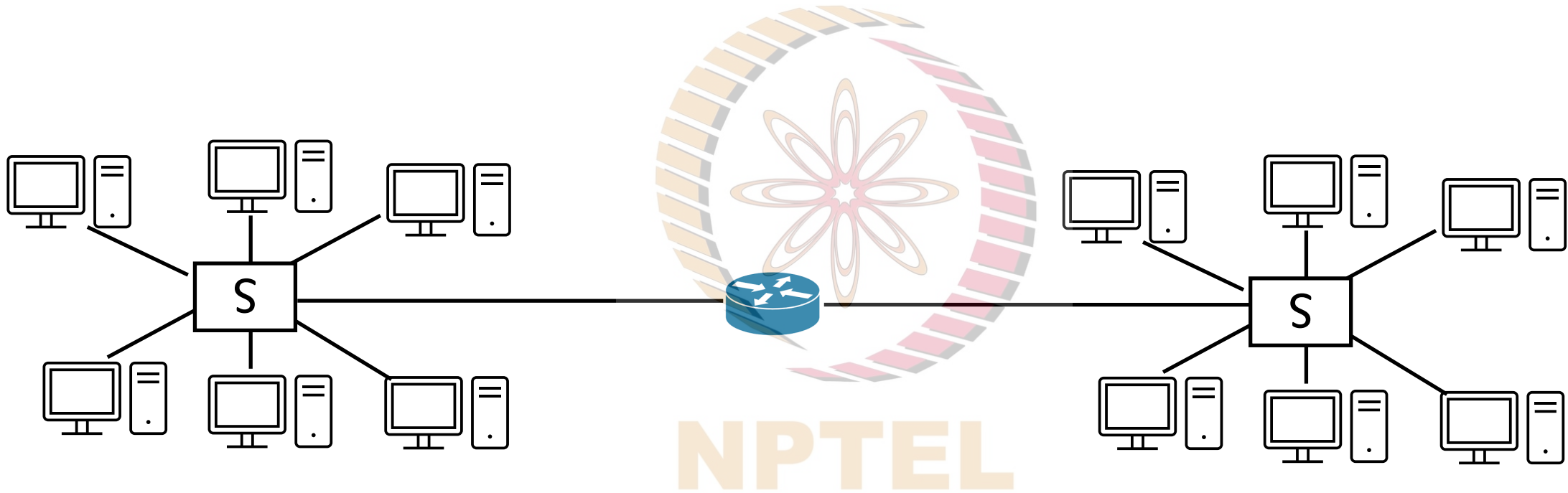
□ Ring



Connecting N users: *Indirectly*



Connecting N Computers: *Indirectly*



Inter-Networks: *Networks of Networks*

